

Adaptive Funnel Research Roadmap

From rule-based trading exits to regime-aware stochastic control

Phase 7 — Regime HMM

Background: Markets switch between hidden regimes such as bull, bear, calm, and crisis.

Definition: Introduce a latent state $R_t \in \{1, \dots, K\}$ evolving as a Markov chain.

Intuition: Use posterior regime probabilities to widen the funnel in unstable regimes and tighten it in stable regimes.

Formulation:
 $r_t | R_t = j \sim \mathcal{N}(\mu_j, \sigma_j^2)$
 $P(R_t = j | r_{1:t})$
 $k_t = \sum_j z_{\alpha,j} w_{j,t} \quad \delta_t = \sum_j \bar{z}_{\alpha,j} \sigma_j^2$

Solution: Estimate $(\mu_j, \sigma_j^2, A_{ij})$ by Baum-Welch / EM on the training data; use forward filtering online on the test fold.

Implementation: Fit HMM on each walk-forward training window, infer regime probabilities live, and set regime-weighted drift / volatility / thresholds into the funnel.

★ *Key idea: HMM is a discrete-state Bayesian filter that complements the Kalman filter.*

Phase 9 — Generalized Momentum

Background: A fixed momentum threshold may be too rigid across different holding times.

Definition: Generalize the sell rule by replacing the fixed condition $M_t \leq 0$ with a time-dependent threshold.

Intuition: Early in the trade, require strongly negative momentum to sell; later, even mild negative momentum can trigger profit-taking.

Formulation:
 $M_t \leq f(t - \tau_b)$
 $f(s) = -\frac{c}{\sqrt{s}}, \quad s = t - \tau_b, \quad c > 0$
Sell if $Z_t > k_t, M_t \leq -\frac{c}{\sqrt{t - \tau_b}}, \Pi_t > 0$

Solution: Choose c by validation or walk-forward tuning; compare against the baseline threshold $M_t \leq 0$.

Implementation: Add c as a tunable parameter, backtest the time-decaying threshold, and compare holding time, Sharpe, and drawdown against the baseline.

★ *Elegant symmetry: funnel width grows like $\sqrt{t - \tau_b}$, while momentum tolerance relaxes like $1/\sqrt{t - \tau_b}$.*

Current Baseline: Adaptive Funnel V1

Background: A trading-exit framework that avoids voluntary realized loss on completed algorithmic exits.

Core objects: X_t = log-price, C_t = predictive center, V_t = cumulative predictive variance

Normalized deviation: $Z_t = \frac{X_t - C_t}{\sqrt{V_t}}$

Profit buffer: $\Pi_t = X_t - X_b - \ell_c, \ell_c = \log\left(\frac{1 + \alpha_c}{1 - \alpha_c}\right)$

Sell rule: Sell when $Z_t > k_b$, momentum weakens, and $\Pi_t > 0$

Intuition: Take profit only when price is sufficiently above model expectation, continuation weakens, and post-cost profit is positive.

🏆 *Goal: keep the no-voluntary-realized-loss guarantee while making the strategy smarter and more optimal.*

Phase 8 — Randomized Stopping

Background: Hard thresholds can be brittle and overly sensitive near the boundary.

Definition: Replace the deterministic sell trigger with a probability of selling.

Intuition: Sell softly near the boundary; recover the option in case a winning case.

Formulation:
 $h_t = \sigma(\alpha(Z_t - k_t) - \delta(\Pi_t + \tau\Pi_t))$
 $D_t \sim \text{Bernoulli}(h_t)$
Sell when $D_t = 1$, while enforcing $\Pi_t > 0$.
As $\alpha \rightarrow \infty$, logistic policy $\rightarrow$ deterministic rule.

Solution: Choose policy parameters by maximizing utility or balancing reward, holding time, and risk.

Implementation: Run Monte Carlo backtests, compare deterministic vs probabilistic exits, and always enforce $\Pi_t > 0$ to preserve the guarantee.

★ *Extra lens: hazard rate and survival analysis for holding-time distribution.*

Phase 10 — HJB / Free-Boundary Theory

Background: Provide a stochastic-control foundation for the adaptive funnel.

Definition: Formulate selling as a constrained optimal stopping problem with transaction costs and holding-time penalty.

Intuition: The adaptive funnel acts as a tractable approximation to an optimal free boundary.

Formulation:
 $V(t, x) = \sup_{\tau \in T} E_t[e^{-r\tau} G(X_{t+\tau}) - \Lambda \int_t^{t+\tau} e^{-\rho s} ds]$
Variational inequality: $\min\{-\mathcal{L}V - \Lambda, V - G\} = 0$
 $\mathcal{L} = \frac{\partial}{\partial t} + \mu \frac{\partial}{\partial x} + \frac{1}{2} \sigma^2 \frac{\partial^2}{\partial x^2}$

Solution: Solve the finite-horizon variational inequality numerically; approximate solution via semi-claims.

Implementation: Use the finite-difference, policy iteration, or penalty methods; compare the computed free boundary with the adaptive funnel boundary; benchmark in-the-money vs out-of-and in-the-entry.

★ *Capstone insight: the free-boundary is the interpretable aim of the adaptive optimal boundary.*

Implementation Order

Phase 9

Phase 7

Phase 8

Phase 10

Theory Ladder



State estimation



latent regimes



probabilistic stopping



stochastic control

Unified Vision

Adaptive Funnel V2/V3 = state estimation + regime awareness + continuation scoring + constrained stopping + optimal-control interpretation

This roadmap showcases mathematics, statistics, stochastic processes, and implementation maturity.